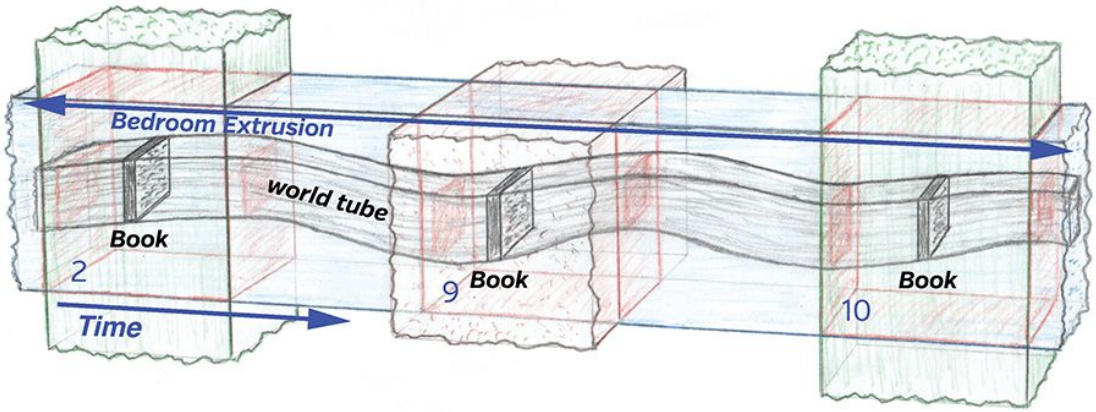


Fig. 30.2. The world tube of a book, within an extrusion of Murph's bedroom. The book and its world tube are drawn much larger than they actually are. *[My own hand sketch.]*

Each object in the bedroom, for example each book, contributes to the bedroom's extrusion. In fact, the book has its own extrusion, which travels forward in time along the blue-arrow direction as part of the bedroom's larger extrusion. We physicists call a variant of this extrusion the book's "world tube." And we call the extrusion of each particle of matter in the book the particle's "world line." So the book's world tube is a bundle of world lines of all the particles that make up the book. Chris and Paul also use this language. The thin lines that you see in the movie, running along the extrusions, are world lines of particles of matter in Murph's bedroom.

In Figure 30.1, Cooper slams his fist on the book's world tube over and over again, creating a gravitational force, which travels backward in time to the moment in Murph's bedroom that he is seeing and then pushes on the book's world tube. The book's tube responds by moving. The tube's motion appears to Cooper as an instantaneous response to his pushes. And the motion becomes a wave traveling leftward down the tube (Figure 30.2).[56] When the motion gets strong enough, the book falls out of the bookcase.

By the time Cooper has received the quantum data from TARS, he has mastered this means of communication. In the movie we see him